TANVIR AHMED

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RESEARCH INTERESTS

Privacy-Preserving Wireless Sensing, Signal Processing, Applied Cryptography, AI/ML.

EDUCATION

Cornell University NYC, US
Ph.D. in Information Science (GPA: **4.18**/4.3) Aug 2023 - Aug 2028 *(Expected)*

- Research: Designing novel and secure wireless sensing systems for next-generation (6G) networks with applied cryptography and normative privacy frameworks. Collaborators: Thijs Roumen, Helen Nissenbaum.
- Committee: Rajalakshmi Nandakumar (Chairperson), Deborah Estrin, and Noah Snavely.
- Digital Life Initiative Doctoral Fellow and PiTech Impact Fellow at Cornell Tech.

Bangladesh University of Engineering and Technology (BUET) Dhaka, Bangladesh
M.Sc. in Electrical & Electronic Engineering (GPA: **3.83**/4.00) Apr 2019 - Jul 2023

- Thesis: Image Super-Resolution Using Wavelet Residual CNNs [online]. Committee: S. M. Mahbubur Rahman (Chairperson), Shaikh Anowarul Fattah, Mohammed Imamul Hassan Bhuiyan, Md. Hasanul Kabir, and Md. Aynal Haque.

Bangladesh University of Engineering and Technology (BUET) Dhaka, Bangladesh
B.Sc. in Electrical & Electronic Engineering (GPA: **3.82**/4.00) Feb 2015 - Apr 2019

- Thesis: Traffic Sign Detection in Streaming Videos. Advisor: S. M. Mahbubur Rahman.

PUBLICATIONS[†]

[C5] mmFHE: mmWave Sensing with End-to-End Fully Homomorphic Encryption
Ahmed, T., Gao, Y., Armouti, A., & Nandakumar, R. [arXiv:2603.22437]

- In Proc. of the *32nd Annual International Conference on Mobile Computing and Networking*, October 26-30, 2026, Austin, Texas, USA. (ACM MobiCom'26)

[P2] PrivyWave: Privacy-Aware Wireless Sensing of Heartbeat
[*]Gao, Y., [*]Ahmed, T., Chang, Z., Roumen, T. & Nandakumar, R. [arXiv:2511.02993]

- Under review.

[P1] SoilSound: Smartphone-based Soil Moisture Estimation
Gao, Y., Ahmed, T., He, S., Cheng, Z. & Nandakumar, R. [arXiv:2509.09823]

- Under review.

[C4] VitalHide: Enabling Privacy-Aware Wireless Sensing of Vital Signs
[*]Gao, Y., [*]Ahmed, T., Chang, Z., Roumen, T. & Nandakumar, R. [open access online]

- In Proc. of the *26th International Workshop on Mobile Computing Systems and Applications*, February 26-27, 2025, California, USA. (ACM HotMobile'25)
- Acceptance rate: 40%.
- News: Poster @ NYC Privacy Day at Google – Fall 2024, Cornell Tech News, Cornell Chronicle, ACM Showcase.

[*]equal contribution. [†]Google Scholar: https://scholar.google.com/citations?user=YVYJl3QAAAAJ&hl=en

[C3] Feasibility of Radio Frequency Based Wireless Sensing of Lead Contamination in Soil
Gao, Y., Ahmed, T., Cheng, Z., Mohammed, M. & Nandakumar, R. [open access online]

- In Proc. of the *21st International Conference on Embedded Wireless Systems and Networks*, December 10-13, 2024, Abu Dhabi, UAE. (EWSN'24)
- Acceptance rate: 22.86%.
- News: Best Paper Award, Cornell Chronicle, Phys.org, MSN, American Technion Society, KARMACTIVE.

[J1] Biomimicry in Nanotechnology: a Comprehensive Review
Himel, M. H., Sikder, B., Ahmed, T. & Choudhury, S. M. [open access online]

- In *Nanoscale Advances* 5, no. 3 (2023): 596-614.

[C2] COVID-19 Identification From Lung CT Scans in a Low-Resource Setting Using a Regularized 3D Convolutional Neural Network
Ahmed, T., Nakib, M., Haque, M. A., & Miah, M. M. M. [pre-print] | [IEEE Xplore library]

- In Proc. of the *12th International Conference on Electrical and Computer Engineering*, December 21-23, 2022, Dhaka, Bangladesh. (ICECE)

[C1] Epileptic Seizure Prediction Using Bandpass Filtering and Convolutional Neural Network
Mustaqeem, N., Rahman, T., Priyo, J. F. B. K., Parvez, M. Z., & Ahmed, T. [pre-print] | [Springer library]

- In Proc. of the *1st International Conference on Machine Intelligence and Emerging Technologies*, Spetember 23-25, Noakhali, Bangladesh. (MIET)

ACADEMIC SERVICE

- Reviewer: ACM CHI ['25, '26], ACM/IEEE HRI ['26], ACM DIS ['25, '26], UbiComp/ISWC ['25], IEEE Signal Processing Letters ['25, '26], ACM MobileHCI ['26]
- Topic Chair: IEEE ECCE ['25]
- Student Volunteer: ACM HotMobile ['25]

WORK EXPERIENCE

Environmental Defense Fund NYC, US

- Graduate Intern @ MethaneSAT Data Platform (Manager: Sasha Ayvazov) May 2026 - Aug 2026
- Developed Masked Prediction, a patch-based method for estimating background methane in scenes with spatial gradients. It hides each patch and predicts its background from the surrounding context, preventing plumes within the patch from influencing the estimate. Pooled local predictions provide a scene-level estimate and uncertainty. On synthetic scenes generated from real Jacobians, the method reduced mean absolute error by about 70% over the production baseline (0.89 vs. 2.91 ppb).

Cornell Tech NYC, US

- Graduate Research Assistant @ Wireless Sensing and Mobile Systems Lab Aug 2023 - *Present*
- Graduate Teaching Assistant: ECE 5260/ORIE 5735 Graph-Based Data Science for Networked Systems (Spring 25), CS/INFO 5304 Data Science in the Wild (Spring 24), INFO 5600 AI for Healthcare (Fall 23, 24), INFO 5375 Machine Learning for Health (Fall 26).

Brac University Dhaka, Bangladesh

- Lecturer @ Department of Computer Science and Engineering Jan 2020 - Aug 2023
  Courses Instructed: CSE 460 VLSI Design, CSE 428 Image Processing, CSE 350 Digital Electronics and Pulse Techniques, CSE 251 Electronic Devices and Circuits, CSE 250 Circuits and Electronics.

SCHOOL PROJECTS

- Peer-to-peer computational resource sharing in networked edge devices [report]
- Impact of multi-avatar and camera perspective on self-presence in VR [report]
- Performance analysis of privacy-preserving logistic regression classifiers on the MNIST dataset [report] [code]
- Segmentation of ground-glass opacity from COVID-19-infected lung CT scans [report] [code]
- Sign-language digit classification with explainable AI [report] [code]
- Stage spotlight automation using deep learning and micro-controllers [code]
- Design, implementation & verification of 32-bit MIPS processor [code], 8×8 Booth-encoded multiplier [report]
- Real-time ECG monitoring and disease detection using Arduino, ECG chip, and WiFi module [report]

TECHNICAL SKILLS

Programming Languages:	Python, C, C++, Assembly, Verilog
Machine Learning Libraries:	TensorFlow, Keras, PyTorch, Scikit-learn, MLX
Circuit Design & Simulation:	PSpice, Proteus, Quartus, Cadence
Numerical Analysis:	MATLAB, Numpy, SciPy
Google Workspace:	Docs, Sheets, Slides, Colab
Writing & Presentation:	Microsoft Word, Microsoft PowerPoint, LaTeX
Millimeter wave:	mmWave Studio
USRP:	GNU Radio
Sleep Medicine:	Rem Logic

HONORS & AWARDS

- PiTech Impact Fellowship (full summer support of $20000), Cornell Tech 2026
- Digital Life Initiative (DLI) Conscientious Tech Design Fellowship ($600), Cornell Tech 2026
- Digital Life Initiative (DLI) Doctoral Fellowship ($5000), Cornell Tech 2025-26
- Student Travel Grant ($800), ACM HotMobile '25, California, USA 2025
- Best Paper Award (€1200), EWSN'24, Abu Dhabi, UAE 2024
- Student Travel Grant (€2000), EWSN'24, Abu Dhabi, UAE 2024
- Photo Contest Winner, "Tech Innovation in Frame" category, Cornell Tech, USA 2023
- Bachelor's Degree with Honors (GPA $\geq$ 3.75), BUET 2019
- Dean's List (Academic Year GPA $\geq$ 3.75), BUET 2015, 2016, 2018
- University Merit Award, BUET 2015, 2016, 2017, 2018
- National Idea Competition, Top 10, Ministry of Power, Energy & Mineral Resources, Bangladesh 2017
- National Physics Olympiad, 1st position, St. Joseph Higher Secondary School, Dhaka, Bangladesh 2014
- Bangladesh Physics Olympiad, 7th position, Dhaka Divisional, Bangladesh 2014